

MTH 418a: Inference-I
Assignment No. 5: Methods of Estimation and Rao Blackwell
Theorem

1. **(Review Problem)** Let $X_1, \dots, X_n$ be a random sample from p.m.f./p.d.f. $f_\theta, \theta \in \Theta$. In each of the following cases, find MME and MLE of θ . Also, determine if they are functions of a minimal sufficient statistic:
 - (i) $X_1 \sim N(\theta, \sigma_0^2), \Theta = \mathbb{R}, \sigma_0$ is a known positive constant;
 - (ii) $X_1 \sim N(\mu_0, \theta^2), \Theta = (0, \infty), \mu_0$ is a known real constant;
 - (iii) $X_1 \sim \text{Exp}(\theta, \sigma_0), \Theta = \mathbb{R}, \sigma_0$ is a known positive constant;
 - (iv) $X_1 \sim \text{Exp}(\mu_0, \theta), \Theta = (0, \infty), \mu_0$ is a known real constant;
 - (v) $X_1 \sim U(\theta - \sigma_0, \theta + \sigma_0), \Theta = \mathbb{R}, \sigma_0$ is a known positive constant;
 - (vi) $X_1 \sim U(\mu_0 - \theta, \mu_0 + \theta), \Theta = (0, \infty), \mu_0$ is a known real constant;
 - (vii) $f_\theta(x) = \alpha_0 \frac{x^{\alpha_0-1}}{\theta^{\alpha_0}} I(0 < x < \theta), \Theta = (0, \infty), \alpha_0$ is a known positive constant;
 - (viii) $f_\theta(x) = \alpha_0 \frac{\theta^{\alpha_0}}{x^{\alpha_0+1}} I(x > \theta), \Theta = (0, \infty), \alpha_0$ is a known positive constant;
 - (ix) $X_1 \sim \text{Bin}(m_0, \theta), \Theta = (0, 1), m_0$ is a known positive integer;
 - (x) $X_1 \sim \text{Bin}(\theta, p_0), \Theta = \{1, 2, \dots\}, p_0 \in (0, 1)$ is a known constant;
 - (xi) $X_1 \sim \text{Poisson}(\theta), \Theta = (0, \infty)$;
 - (xii) $X_1 \sim \text{Gamma}(\theta, \alpha_0), \Theta = (0, \infty), \alpha_0$ is a known positive constant;
 - (xiii) $X_1 \sim \text{Gamma}(\sigma_0, \theta), \Theta = (0, \infty), \sigma_0$ is a known positive constant;
 - (xiv) $X_1 \sim \text{DEXP}(\theta, \sigma_0), \Theta = \mathbb{R}, \sigma_0$ is a known positive constant;
 - (xv) $X_1 \sim \text{DEXP}(\mu_0, \theta), \Theta = (0, \infty), \mu_0$ is a known real constant;
2. Find two examples where the MLE is not unique.
3. Let $X \sim \text{Gamma}(1, \alpha), \alpha > 0$. Show that $E(\ln X) = \psi(\alpha)$, where $\psi(\alpha) = \frac{d}{d\alpha} \ln \Gamma(\alpha) = \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}, \alpha > 0$, is called the digamma function. Show that:
 - (i) $\psi(\alpha) < \ln \alpha, \forall \alpha > 0$;
 - (ii) $\psi(\alpha)$ is an increasing function on $(0, \infty)$.
4.
 - (i) Let $X_1, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$, where $\underline{\theta} = (\mu, \sigma) \in \mathbb{R} \times (0, \infty) = \Theta$. Find the MLE and the MME of $\tau(\underline{\theta}) = P_{\underline{\theta}}(X_1 > 1)$;
 - (ii) Let $X_1, \dots, X_n$ be a random sample from $\text{Gamma}(\sigma, \alpha)$, where $\underline{\theta} = (\sigma, \alpha) \in (0, \infty) \times (0, \infty) = \Theta$. Find the MLEs and MMEs of $\underline{\theta}$ and $\tau(\underline{\theta}) = \text{Var}_{\underline{\theta}}(X_1)$;
 - (iii) Let $X_1, \dots, X_n$ be a random sample from $\text{DEXP}(\mu, \sigma), \underline{\theta} = (\mu, \sigma), \Theta = \mathbb{R} \times (0, \infty)$. Find the MLE and the MME of $\underline{\theta}$.

5. Let X_1, X_2 be a random sample from a p.d.f. $f_\theta(x) = \frac{2}{\theta^2}(\theta - x)I(0 < x < \theta), \theta \in (0, \infty) = \Theta$. Find the MME and the MLE of θ . Are they functions of a minimal sufficient statistic?
6. Let $X_1, \dots, X_n$ be a random sample from p.m.f./p.d.f. $f_\theta, \theta \in \Theta$. In each of the following cases, find MME and MLE of θ . Also determine if they are functions of a minimal sufficient statistic:
 - (i) $X_1 \sim N(\theta, \sigma_0^2), \Theta = (a, b), a, b$ ($a < b$) and σ_0 are known positive constants;
 - (ii) $X_1 \sim \text{Exp}(\theta, \sigma_0), \Theta = (a, b), a, b$ ($a < b$) and σ_0 are known positive constants;
 - (iii) $X_1 \sim \text{Bin}(m_0, \theta), \Theta = [\frac{1}{2}, 1), m_0$ is a known positive integer;
 - (iv) $X_1 \sim U(\theta_1, \theta_2), \underline{\theta} = (\theta_1, \theta_2), \Theta = \{(x, y) \in \mathbb{R}^2 : x < y\}$;
 - (v) $X_1 \sim U(\theta_1 - \theta_2, \theta_1 + \theta_2), \underline{\theta} = (\theta_1, \theta_2), \Theta = \{(x, y) \in \mathbb{R}^2 : -\infty < x < \infty, y > 0\}$.
7. Let $X_1, \dots, X_n$ be a random sample from $\text{Exp}(0, \theta), \theta \in \Theta = (0, \infty)$. Let c be a positive constant. What is being observed is only those X_i s whose values are less than c . Suppose that the observed values are $Y_1, \dots, Y_m$ and remaining $(n - m)$ observations that exceed c are not observed. Find the MLE of θ .
8. Prove the following inequalities for any non-degenerate random variable X : (i) $E(e^X) > e^{E(X)}$; (ii) $E(X)E(\frac{1}{X}) > 1$, provided $P(X > 0) = 1$; (iii) $E(\ln X) < \ln E(X)$, provided $P(X > 0) = 1$. Hence prove the AM-GM-HM inequality

$$\sum_{i=1}^n a_i w_i > \prod_{i=1}^n a_i^{w_i} > \frac{1}{\sum_{i=1}^n \frac{w_i}{a_i}},$$

for any positive constants $w_1, \dots, w_n, a_1, \dots, a_n$, such that $\sum_{i=1}^n w_i = 1$ and not all a_i s are the same.

9. Let $X_1, \dots, X_n$ be a random sample from $N(\theta, 1), \theta \in \mathbb{R} = \Theta$. For estimating θ under a loss function, consider the randomized estimator $\delta(\underline{x}) \sim N(\frac{x_1 + x_2}{2}, 1)$.
 - (i) Find a randomized estimator that is based on a minimal sufficient statistic and has the same risk function as $\delta(\underline{x})$;
 - (ii) Under the squared error loss function find a non-randomized estimator better than δ .
10. Let $X_1, \dots, X_n$ ($n \geq 2$) be a random sample from $\text{Exp}(0, \theta), \theta \in (0, \infty) = \Theta$. For estimating θ under a loss function, consider the randomized estimator $\delta(\underline{x}) \sim U(\frac{n-1}{n}\bar{x}_{n-1}, \bar{x}_{n-1})$, where $\bar{x}_{n-1} = \frac{1}{n-1} \sum_{i=1}^{n-1} x_i$.
 - (i) Find a randomized estimator based on a minimal sufficient statistic having the same risk function as $\delta(\underline{x})$;
 - (ii) Under the squared error loss function find a non-randomized estimator better than δ .

11. Let $X_1, \dots, X_n$ ($n \geq 2$) be a random sample from $N(\theta, 1)$, where $\theta \in \mathbb{R} = \Theta$ is the unknown parameter. For estimating θ , suppose that the loss function is either the absolute error loss function or the squared error loss function.
 - (i) Find an estimator dominating the estimator $\delta_1(\underline{X}) = \sum_{i=1}^n \alpha_i X_i$, where $\alpha_1, \dots, \alpha_n$ are positive constants such that $\sum_{i=1}^n \alpha_i = 1$ and not all α_i s are the same;
 - (ii) Let $n = 2m + 1$. Find a randomized estimator, based on $\overline{X}$, which has the same risk function as the sample median $\delta_2(\underline{X}) = X_{(m+1)}$;
 - (iii) Find an estimator better than $\delta_2(\underline{X}) = X_{(m+1)}$.
12. Let $X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1) = \Theta$. For estimating θ under the squared error loss function, consider the randomized estimator δ , such that $P(\delta(x) = \frac{x}{n}) = P(\delta(x) = \frac{1}{2}) = \frac{1}{2}$. Find a nonrandomized estimator better than δ . Also calculate the risk functions of the two estimators.
13. Let $X_1, \dots, X_n$ be a random sample from $U(0, \theta)$, where $\theta \in (0, \infty) = \Theta$ is the unknown estimator. Consider estimation of θ under the absolute error loss function.
 - (i) Find an estimator, based on a minimal sufficient statistics, that is better than $\delta_0(\underline{X}) = \overline{X}$;
 - (ii) Find an estimator, based on a minimal sufficient statistics, that is better than $\delta_1(\underline{X}) = X_{(n-1)}$.

MTH 418 a: Inference - I

Assignment No. 5: Method of Estimation and Rao-Blackwell Theorem

Solution

Problem No. 1

For fixed data point $\underline{x} = (x_1, \dots, x_n)$

$$(i) \ln L_{\underline{x}}(\theta) = -\frac{n}{2} \ln(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \theta)^2, \quad -\infty < \theta < \infty$$

$$\frac{\partial}{\partial \theta} \ln L_{\underline{x}}(\theta) = \frac{1}{\sigma^2} n(\bar{x} - \theta)$$

$$L_{\underline{x}}(\theta) \uparrow \text{ in } \theta \in (-\infty, \bar{x}), \quad \downarrow \text{ in } \theta \in (\bar{x}, \infty)$$

$$\Rightarrow \hat{\theta}_{MLE} = \bar{x}$$

$$E_{\theta}(x_i) = \theta \Rightarrow \hat{\theta}_{MNE} = \bar{x}$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MNE}$ are functions of minimal sufficient statistic $\bar{x}$

$$(ii) \ln L_{\underline{x}}(\theta) = -\frac{n}{2} \ln(2\pi) - n \ln \theta - \frac{1}{2\theta^2} \sum_{i=1}^n (x_i - \mu_0)^2, \quad \theta > 0$$

$$\frac{\partial}{\partial \theta} \ln L_{\underline{x}}(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^3} \sum_{i=1}^n (x_i - \mu_0)^2$$

$$L_{\underline{x}}(\theta) \uparrow \text{ in } \theta \in (0, \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)^2}), \text{ and } \downarrow \text{ in } \theta \in (\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)^2}, \infty)$$

$$\Rightarrow \hat{\theta}_{MLE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)^2}$$

$$E_{\theta}(x_i) = \mu_0 \rightarrow \text{Can not be used to find MNE}$$

$$E_{\theta}(x_i^2) = \theta^2 + \mu_0^2 \Rightarrow \hat{\theta}_{MNE}^2 + \mu_0^2 = \frac{1}{n} \sum_{i=1}^n x_i^2$$

$$\Rightarrow \hat{\theta}_{MNE} = \sqrt{\frac{1}{n} \sum_{i=1}^n x_i^2 - \mu_0^2}$$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic $T_1 = \sum_{i=1}^n (x_i - \mu_0)^2$
 $\hat{\theta}_{MNE}$ is not a function of $\hat{\theta}_{MLE}$
 $\frac{1}{n} \sum_{i=1}^n x_i^2$ is not a function of $\frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)^2$ (minimal sufficient)
 $\frac{1}{n} \sum_{i=1}^n x_i^2$ is not a function of $\frac{1}{n} \sum_{i=1}^n x_i^2$

$$(III) \quad \ln L_2(\theta) = -n \ln \sigma_0 - \frac{1}{\sigma_0} \sum_{i=1}^n x_i + \frac{n\theta}{\sigma_0}, \quad -\sigma_0 < \theta < x_{(n)}$$

is increasing in $\theta \in (-\sigma_0, x_{(n)})$

$$\Rightarrow \hat{\theta}_{MLE} = x_{(n)}$$

$$E_\theta(x_1) = \theta + \sigma_0 \Rightarrow \hat{\theta}_{MLE} + \sigma_0 = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n (x_i - \sigma_0)$$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic $x_{(n)}$

$\hat{\theta}_{MLE}$ is not a function of minimal sufficient statistic $x_{(n)}$ as the minimal sufficient statistic is not a function of $\hat{\theta}_{MLE}$.

$$(IV) \quad \ln L_2(\theta) = -n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n (x_i - \mu_0), \quad \theta > 0$$

$$L_2(\theta) \text{ is } \frac{d}{d\theta} \ln L_2(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n (x_i - \mu_0) \text{ increasing in } \theta \in (0, \frac{1}{n} \sum_{i=1}^n (x_i - \mu_0) + \mu_0) \text{ and } \downarrow \text{ in } (\frac{1}{n} \sum_{i=1}^n (x_i - \mu_0) + \mu_0, \infty)$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)$$

$$E_\theta(x_1) = \mu_0 + \theta \Rightarrow \mu_0 + \hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n (x_i - \mu_0)$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MLE}$ are functions of minimal sufficient statistic $T = \sum_{i=1}^n (x_i - \mu_0)$.

$$(V) \quad L_2(\theta) = \frac{1}{(2\sigma_0)^n}, \quad x_{(n)} - \sigma_0 \leq \theta \leq x_{(1)} + \sigma_0$$

is a constant function of $\theta \in [x_{(n)} - \sigma_0, x_{(1)} + \sigma_0]$. Thus any $\hat{\theta}(x)$, such that $P_0(x_{(n)} - \sigma_0 \leq \hat{\theta}(x) \leq x_{(1)} + \sigma_0) = 1, \forall \theta \in \mathbb{R}$ is an MLE. In particular $\hat{\theta}_{1,MLE} = x_{(n)} - \sigma_0, \hat{\theta}_{2,MLE} = x_{(1)} + \sigma_0, \hat{\theta}_{3,MLE,\alpha} = \alpha(x_{(n)} - \sigma_0) + (1-\alpha)(x_{(1)} - \sigma_0), 0 \leq \alpha \leq 1$, all are MLEs

$$E_{\theta}(X_1) = \theta \Rightarrow \hat{\theta}_{MLE} = \bar{X}$$

$\hat{\theta}_{MLE}$ are functions of minimal sufficient statistic $T = (X_{(1)}, X_{(n)})$
 $\hat{\theta}_{MLE} = \bar{X}$ is not a function of minimal sufficient statistic as minimal sufficient statistic $T = (X_{(1)}, X_{(n)})$ is not a function of $\bar{X}$.

$$(V) \quad L_{\lambda}(\theta) = \frac{1}{(2\theta)^n}, \quad \text{if } 0 > \max\{|x_1 - \mu_0|, \dots, |x_n - \mu_0|\} = T(\lambda), \quad \lambda \in [a, \infty)$$

is a decreasing function of $\theta \in [T(\lambda), \infty)$

$$\Rightarrow \hat{\theta}_{MLE} = T(\lambda) = \max\{|x_1 - \mu_0|, \dots, |x_n - \mu_0|\}$$

$E_{\theta}(X_1) = \mu_0 \rightarrow$ Cannot be used to find MLE

$$E_{\theta}(X_1^2) = \frac{\theta^2}{3} + \mu_0^2 \Rightarrow \frac{\hat{\theta}_{MLE}^2}{3} + \mu_0^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

$$\Rightarrow \hat{\theta}_{MLE} = \sqrt{\frac{3}{n} \sum_{i=1}^n (X_i^2 - \mu_0^2)}$$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic T

$\hat{\theta}_{MLE}$ is not a function of minimal sufficient statistic T .

$$(VI) \quad \ln L_{\lambda}(\theta) = n \ln \alpha_0 - n \alpha_0 \ln \theta + (\alpha_0 - 1) \sum_{i=1}^n \ln x_i, \quad 0 > x_{(n)}$$

is a $\downarrow$ function of $\theta \in [x_{(n)}, \infty)$

$$\Rightarrow \hat{\theta}_{MLE} = x_{(n)}$$

$$E_{\theta}(X_1) = \frac{\alpha_0}{\alpha_0 + 1} \theta \Rightarrow \frac{\alpha_0}{\alpha_0 + 1} \hat{\theta}_{MLE} = \bar{X} \Rightarrow \hat{\theta}_{MLE} = \frac{\alpha_0 + 1}{\alpha_0} \bar{X}$$

$\hat{\theta}_{MLE} = x_{(n)}$ is a function of minimal sufficient statistic $x_{(n)}$

$\hat{\theta}_{MLE} = \frac{\alpha_0 + 1}{\alpha_0} \bar{X}$ is not a function of a minimal sufficient statistic as $\bar{X}$ is not a function of minimal sufficient statistic $x_{(n)}$.

(VIII) $\ln L_2(\theta) = n \ln \alpha_0 + n \alpha_0 \ln \theta - (\alpha_0 + 1) \sum_{i=1}^n \ln x_i$, $-\alpha_0 < \theta < x_{(1)}$
 is an $\uparrow$ function of $\theta \in (-\alpha_0, x_{(1)})$

$$\Rightarrow \hat{\theta}_{MLE} = x_{(1)}$$

For $\alpha_0 > 1$

$$E_0(x_i) = \frac{\alpha_0}{\alpha_0 - 1} \theta \Rightarrow \frac{\alpha_0}{\alpha_0 - 1} \hat{\theta}_{MLE} = \bar{X} \Rightarrow \hat{\theta}_{MLE} = \frac{\alpha_0 - 1}{\alpha_0} \bar{X}$$

Note that, for $0 < \alpha_0 \leq 1$, MLE does not exist.

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic $T = x_{(1)}$
 $\hat{\theta}_{MLE}$ is not a function of a minimal sufficient statistic.

(IX) $\ln L_2(\theta) = \ln \left(\prod_{i=1}^n \binom{m_0}{x_i} \right) + \left(\sum_{i=1}^n x_i \right) \ln \theta + (n - \sum_{i=1}^n x_i) \ln(1 - \theta)$, $0 < \theta < 1$

$$\frac{\partial}{\partial \theta} \ln L_2(\theta) = \frac{n \bar{x}}{\theta} - \frac{n(m_0 - \bar{x})}{1 - \theta}, \quad 0 < \theta < 1$$

$L_2(\theta)$ is increasing in $\theta \in (0, \frac{\bar{x}}{m_0})$ and $\downarrow$ in $\theta \in (\frac{\bar{x}}{m_0}, 1)$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{m_0}$$

$$E_0(x_i) = m_0 \theta \Rightarrow m_0 \hat{\theta}_{MLE} = \bar{x} \Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{m_0}$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MLE}$ are functions of minimal sufficient statistic $T = \sum_{i=1}^n x_i$.

(X) $L_2(\theta) = \prod_{i=1}^n \left\{ \binom{\theta}{x_i} p_0^{x_i} (1-p_0)^{\theta-x_i} \right\}$, $\theta \in t_1, t_1+1, \dots$,

Where $t_1 = \max \{x_1, x_2, \dots, x_n\}$.

$$\frac{L_2(\theta)}{L_2(\theta-1)} = (1-p_0)^n \prod_{i=1}^n \left(1 - \frac{x_i}{\theta}\right)^{-1}, \quad \theta = t_1, t_1+1, \dots$$

$$\frac{L_2(\theta)}{L_2(\theta-1)} \geq 1 \Leftrightarrow L_2(\theta) \geq L_2(\theta-1) \Leftrightarrow (1-p_0)^n \prod_{i=1}^n \left(1 - \frac{x_i}{\theta}\right)^{-1} \geq 1$$

$$\frac{L_2(\theta)}{L_2(\theta-1)} \leq 1 \Leftrightarrow L_2(\theta) \leq L_2(\theta-1) \Leftrightarrow (1-p_0)^n \prod_{i=1}^n \left(1 - \frac{x_i}{\theta}\right)^{-1} \leq 1$$

$$\text{Let } \psi(\theta) = (1-p_0)^n \prod_{i=1}^n \left(1 - \frac{x_i}{\theta}\right)^{-1}, \quad \theta = t_1, t_2, \dots$$

Considering θ as a continuous variable we conclude that $\psi(\theta)$ is a ^{strictly} decreasing function of θ on (t_1, ∞) , taking all values in the interval $((1-p_0)^n, \infty)$. Thus there is a unique $\hat{\theta}$ such that $\psi(\hat{\theta}) = 1$,

$L_X(\theta)$ is $\uparrow$ in $\theta \in (t_1, \hat{\theta})$, $\downarrow$ in $(\hat{\theta}, \infty)$. Thus

$$\hat{\theta}_{MLE} = \begin{cases} \hat{\theta}, & \text{if } \hat{\theta} \text{ is an integer} \\ \arg \max \{L_X(\lceil \hat{\theta} \rceil), L_X(\lceil \hat{\theta} \rceil + 1)\}, & \text{otherwise} \end{cases}$$

where $\lceil \hat{\theta} \rceil$ = largest integer not exceeding $\hat{\theta}$.

$$E_0(x_i) = \theta p_0 \Rightarrow \hat{\theta}_{MLE} p_0 = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{n p_0} \sum_{i=1}^n x_i = \frac{\bar{X}}{p_0} \rightarrow \text{may not be an integer}$$

$$\text{We may take } \hat{\theta}_{MLE} = \left\lfloor \frac{\bar{X}}{p_0} \right\rfloor.$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MNE}$ are functions of minimal sufficient statistic $T = (X_{(1)}, \dots, X_{(n)})$.

$$(x1) \quad \ln L_2(\theta) = -n\theta + n\bar{x} \ln \theta - \ln \left(\prod_{i=1}^n \frac{1}{\theta} \right), \quad \theta > 0$$

$$\frac{\partial \ln L_2(\theta)}{\partial \theta} = -n + \frac{n\bar{x}}{\theta}, \quad \theta > 0$$

$L_2(\theta)$ is $\uparrow$ in $\theta \in (0, \bar{x}]$ and $\downarrow$ in $\theta \in (\bar{x}, \infty)$

$$\Rightarrow \hat{\theta}_{MLE} = \bar{x}$$

$$E_0(x_i) = \theta \Rightarrow \hat{\theta}_{MNE} = \bar{x}$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MNE}$ are functions of minimal sufficient

Statistic $T = \sum_{i=1}^n x_i$

$$(XII) \quad \ln L_2(\theta) = -n \ln \Gamma(\alpha_0) - n \alpha_0 \ln \theta + (\alpha_0 - 1) \sum_{i=1}^n \ln x_i - \frac{n \bar{x}}{\theta}, \quad \theta > 0$$

$$\frac{\partial \ln L_2(\theta)}{\partial \theta} = -\frac{n \alpha_0}{\theta} + \frac{n \bar{x}}{\theta^2}, \quad \theta > 0$$

$L_2(\theta)$ is $\uparrow$ in $\theta \in (0, \frac{\bar{x}}{\alpha_0})$ and $\downarrow$ in $\theta \in (\frac{\bar{x}}{\alpha_0}, \infty)$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{\alpha_0}$$

$$E_0(X_1) = \alpha_0 \theta \Rightarrow \alpha_0 \hat{\theta}_{MLE} = \bar{x} \Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{\alpha_0}$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MNE}$ are functions of minimal sufficient statistic

$$T(X) = \sum_{i=1}^n X_i$$

$$(XIII) \quad \ln L_2(\theta) = -n \ln \Gamma(\theta) - n \theta \ln \sigma_0 + (\theta - 1) \sum_{i=1}^n \ln x_i - \frac{n \bar{x}}{\sigma_0}, \quad \theta > 0$$

$$\frac{\partial}{\partial \theta} \ln L_2(\theta) = -n \psi(\theta) - n \ln \sigma_0 + \sum_{i=1}^n \ln x_i, \quad \theta > 0 \quad (\psi(\theta) = \frac{d}{d\theta} \ln \Gamma(\theta) = \frac{\Gamma'(\theta)}{\Gamma(\theta)})$$

$$\frac{\partial^2}{\partial \theta^2} \ln L_2(\theta) = -n \psi'(\theta) < 0, \quad \forall \theta \in (0, \infty) \quad (\text{See Problem 3 (iii)})$$

$$\begin{aligned} \text{Then} \quad \hat{\theta}_{MLE} &= \psi^{-1} \left(\frac{1}{n} \sum_{i=1}^n \ln x_i - \ln \sigma_0 \right) \\ &= \psi^{-1} \left(\frac{1}{n} \sum_{i=1}^n \ln \frac{x_i}{\sigma_0} \right), \end{aligned}$$

$$\text{Where } \psi(\theta) = \frac{\Gamma'(\theta)}{\Gamma(\theta)} = \frac{d}{d\theta} \ln \Gamma(\theta), \quad \theta > 0.$$

$$E_0(X_1) = \sigma_0 \theta \Rightarrow \sigma_0 \hat{\theta}_{MLE} = \bar{x} \Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{\sigma_0}$$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic

$$T = \ln \left(\prod_{i=1}^n X_i \right) = \sum_{i=1}^n \ln X_i$$

$\hat{\theta}_{MNE}$ is not a function of minimal sufficient statistic.

(XIV) $\ln L_2(\theta) = -n \ln(2\sigma) - \frac{1}{\sigma} \sum_{i=1}^n |x_i - \theta|$, $-\infty < \theta < \infty$
 is maximized at $\hat{\theta} = x_m = \text{median}\{x_1, \dots, x_n\}$

$$= \begin{cases} x_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd} \\ \hat{\theta}(x), & \text{if } n \text{ is even} \end{cases}$$

Where $P_0(\hat{\theta}(x) \in [x_{(\frac{n}{2})}, x_{(\frac{n}{2}+1)}]) = 1$, $\forall \theta \in \Theta$

$\Rightarrow \hat{\theta}_{MLE} = \begin{cases} x_{(\frac{n+1}{2})}, & \text{if } n \text{ is odd} \\ \hat{\theta}(x), & \text{if } n \text{ is even} \end{cases}$

Where $\hat{\theta}(x)$ is such that $P_0(\hat{\theta}(x) \in [x_{(\frac{n}{2})}, x_{(\frac{n}{2}+1)}]) = 1$,
 $\forall \theta \in \Theta$. Note that here the MLE may not be unique if
 n is even.

$E_0(x_i) = \theta \Rightarrow \hat{\theta}_{MLE} = \bar{x}$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient Statistic $T = (x_{(1)}, \dots, x_{(n)})$

$\hat{\theta}_{MLE}$ is also a function of minimal sufficient Statistic $T = (x_{(1)}, \dots, x_{(n)})$

(XV) $\ln L_2(\theta) = -n \ln 2 - n \ln \theta - \frac{1}{\theta} \sum_{i=1}^n |x_i - \mu_0|$

$\frac{\partial}{\partial \theta} \ln L_2(\theta) = -\frac{n}{\theta} + \frac{1}{\theta^2} \sum_{i=1}^n |x_i - \mu_0|$

$L_2(\theta)$ is $\uparrow$ as $\theta \in (-\infty, \frac{1}{n} \sum_{i=1}^n |x_i - \mu_0|)$ and $\downarrow$ as $\theta \in (\frac{1}{n} \sum_{i=1}^n |x_i - \mu_0|, \infty)$

$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{n} \sum_{i=1}^n |x_i - \mu_0|$

$E_0(x_i) = \mu_0 \rightarrow$ Cannot be used to find MLE

$E_0(x_i^2) = 2\theta^2 + \mu_0^2$ | $Z = \frac{|x_i - \mu_0|}{\theta} \sim \text{Exp}(0, 1)$

$\Rightarrow 2\hat{\theta}_{MLE}^2 + \mu_0^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 \Rightarrow \hat{\theta}_{MLE} = \sqrt{\frac{1}{2n} \sum_{i=1}^n (x_i^2 - \mu_0^2)}$

$\rightarrow$ may not exist as one
 may have $\sum_{i=1}^n x_i^2 < n\mu_0^2$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic

$$T = \sum_{i=1}^n |x_i - \mu|$$

$\hat{\theta}_{MLE}$ is not a function of minimal sufficient statistic.

Problem 2 Example 1: See Problem 1 (v)

Example 2: See Problem 1 (xiv)

Example 3: Let x_1, x_2 be a random sample from the Cauchy distribution having p.d.f.

$$f_0(x) = \frac{1}{\pi} \cdot \frac{1}{1+(x-\theta)^2}, \quad -\infty < x < \infty, \quad -\infty < \theta < \infty.$$

For given sample points $\underline{x} = (x_1, x_2) \in \mathbb{R}^2$, the log-likelihood function is

$$\ln L_{\underline{x}}(\theta) = -2 \ln \pi - \ln(1+(x_1-\theta)^2) - \ln(1+(x_2-\theta)^2), \quad -\infty < \theta < \infty$$

$$\begin{aligned} \frac{\partial}{\partial \theta} \ln L_{\underline{x}}(\theta) &= \frac{2(x_1-\theta)}{1+(x_1-\theta)^2} + \frac{2(x_2-\theta)}{1+(x_2-\theta)^2} \\ &= \frac{(x_1+x_2-2\theta)(1+(x_1-\theta)(x_2-\theta))}{(1+(x_1-\theta)^2)(1+(x_2-\theta)^2)} \\ &= \frac{(x_1+x_2-2\theta)(\theta^2 - (x_1+x_2)\theta + x_1x_2 + 1)}{(1+(x_1-\theta)^2)(1+(x_2-\theta)^2)} > (<) 0 \end{aligned}$$

$$\Leftrightarrow (x_1+x_2-2\theta)(\theta^2 - (x_1+x_2)\theta + x_1x_2 + 1) > (<) 0$$

Consider the discriminant Δ of quadratic equation $\theta^2 - (x_1+x_2)\theta + x_1x_2 + 1 = 0$

$$\Delta = (x_1+x_2)^2 - 4x_1x_2 - 4 = (x_1-x_2)^2 - 4$$

Case I $|x_1-x_2| < 2$

Here $\theta^2 - (x_1+x_2)\theta + x_1x_2 + 1 > 0, \quad \forall \theta \in \mathbb{R}$

$L_{\underline{x}}(\theta) \uparrow$ as $\theta \in (-\infty, \frac{x_1+x_2}{2})$, $\downarrow$ as $\theta \in (\frac{x_1+x_2}{2}, \infty)$

$$\hat{\theta}_{MLE} = \frac{x_1+x_2}{2}$$

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Case II: $|\lambda_1 - \lambda_2| \geq 2$

$$\text{Let } \hat{\theta}_1 = \frac{\lambda_1 + \lambda_2 - \sqrt{\Delta}}{2}, \quad \hat{\theta}_2 = \frac{\lambda_1 + \lambda_2 + \sqrt{\Delta}}{2},$$

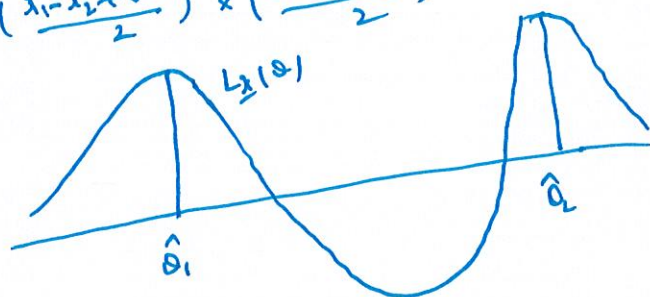
$$\text{And that } L_2(\theta) > 0 \quad \text{if} \quad \theta < \frac{\lambda_1 + \lambda_2 - \sqrt{\Delta}}{2} \quad \text{or} \quad \frac{\lambda_1 + \lambda_2}{2} < \theta < \frac{\lambda_1 + \lambda_2 + \sqrt{\Delta}}{2}$$

$$L_2(\theta) < 0 \quad \text{if} \quad \frac{\lambda_1 + \lambda_2 - \sqrt{\Delta}}{2} < \theta < \frac{\lambda_1 + \lambda_2}{2} \quad \text{or} \quad \theta > \frac{\lambda_1 + \lambda_2 + \sqrt{\Delta}}{2}$$

$$\lim_{\theta \rightarrow \pm\infty} L_2(\theta) = 0$$

$$L_2\left(\frac{\lambda_1 + \lambda_2 - \sqrt{\Delta}}{2}\right) = L_2\left(\frac{\lambda_1 + \lambda_2 + \sqrt{\Delta}}{2}\right) = \frac{1}{\pi} \cdot \frac{1}{\left(\frac{\lambda_1 - \lambda_2 + \sqrt{\Delta}}{2}\right)^2 \times \left(\frac{\lambda_1 - \lambda_2 - \sqrt{\Delta}}{2}\right)^2}$$

Thus for $|\lambda_1 - \lambda_2| \geq 2$, there are two MLEs $\hat{\theta}_1$ and $\hat{\theta}_2$.



Problem No. 3

$$\Gamma(\alpha) = \int_0^{\infty} e^{-x} x^{\alpha-1} dx$$

$$\Gamma'(\alpha) = \frac{d}{d\alpha} \int_0^{\infty} e^{-x} x^{\alpha-1} dx = \int_0^{\infty} \frac{d}{d\alpha} (e^{-x} x^{\alpha-1}) dx$$

$$= \int_0^{\infty} \ln x e^{-x} x^{\alpha-1} dx \quad \dots (I)$$

$$\Rightarrow \psi(\alpha) = \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} = \frac{1}{\Gamma(\alpha)} \int_0^{\infty} \ln x e^{-x} x^{\alpha-1} dx = E(\ln x)$$

(a) $h(x) = -\ln x$, $x > 0$, is a convex function. On using the Jensen inequality we get

$$E(-\ln x) > -\ln E(x)$$

$$\Rightarrow E(\ln x) < \ln x$$

$$\Rightarrow \psi(\alpha) < \ln x, \quad \alpha > 0$$

$$(ii) \quad \psi'(\alpha) = \frac{\Gamma(\alpha) \Gamma''(\alpha) - (\Gamma'(\alpha))^2}{(\Gamma(\alpha))^2}$$

$$= \frac{\Gamma''(\alpha)}{\Gamma(\alpha)} - (\psi(\alpha))^2$$

$$\begin{aligned} \Gamma''(\alpha) &= \frac{d}{d\alpha} \Gamma'(\alpha) = \frac{d}{d\alpha} \int_0^{\infty} \ln x e^{-x} x^{\alpha-1} dx \\ &= \int_0^{\infty} \frac{d}{d\alpha} (\ln x e^{-x} x^{\alpha-1}) dx \\ &= \int_0^{\infty} (\ln x)^2 e^{-x} x^{\alpha-1} dx \end{aligned}$$

$$\begin{aligned} \Rightarrow \psi'(\alpha) &= \frac{1}{\Gamma(\alpha)} \int_0^{\infty} (\ln x)^2 e^{-x} x^{\alpha-1} dx - (E(\ln x))^2 \\ &= E((\ln x)^2) - (E(\ln x))^2 = \text{Var}(\ln x) > 0, \quad \forall \alpha \in (0, \infty) \end{aligned}$$

$\Rightarrow \psi(\alpha)$ is an increasing function of $\alpha \in (0, \infty)$.

For given data point $\underline{x}$,

Problem No. 4

$$(i) \quad \ln L_{\underline{x}}(\underline{\theta}) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2, \quad -\infty < \mu < \infty, \sigma^2 > 0$$

$$\frac{\partial \ln L_{\underline{x}}(\underline{\theta})}{\partial \mu} = \frac{1}{\sigma^2} \sum_{i=1}^n (x_i - \mu)$$

$$\frac{\partial}{\partial \sigma^2} \ln L_{\underline{x}}(\underline{\theta}) = -\frac{n}{2\sigma^2} + \frac{1}{2(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial^2}{\partial \mu^2} \ln L_{\underline{x}}(\underline{\theta}) = -\frac{n}{\sigma^2}, \quad \frac{\partial^2}{\partial (\sigma^2)^2} \ln L_{\underline{x}}(\underline{\theta}) = \frac{n}{2(\sigma^2)^2} - \frac{1}{(\sigma^2)^3} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial^2}{\partial \mu \partial \sigma^2} \ln L_{\underline{x}}(\underline{\theta}) = -\frac{1}{(\sigma^2)^2} \sum_{i=1}^n (x_i - \mu)$$

The critical point is $(\hat{\mu}, \hat{\sigma}^2) = (\bar{x}, \hat{\sigma}_n^2)$, where $\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2$

$$H_{11} = \left[\frac{\partial^2}{\partial \mu^2} \ln L_{\underline{x}}(\underline{\theta}) \right]_{(\hat{\mu}, \hat{\sigma}^2)} = -\frac{n}{\hat{\sigma}^2} < 0, \quad H_{12} = \left[\frac{\partial^2}{\partial \mu \partial \sigma^2} \ln L_{\underline{x}}(\underline{\theta}) \right]_{(\hat{\mu}, \hat{\sigma}^2)} = 0$$

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$$H_{22} = \left(\frac{\partial^2}{\partial \sigma^2} L_2(\underline{\theta}) \right)_{(\hat{\mu}, \hat{\sigma}^2)} = -\frac{n}{2(\hat{\sigma}^2)^2}$$

$$|H| = H_{11} H_{22} - H_{12}^2 = \frac{n^2}{2(\hat{\sigma}^2)^3} > 0$$

$$\Rightarrow (\hat{\mu}_{MLE}, \hat{\sigma}_{MLE}^2) = (\bar{x}, S_n^2), \text{ where } S_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

$$\Rightarrow \text{The MLE of } \gamma(\underline{\theta}) = 1 - \Phi\left(\frac{1-\mu}{\sigma}\right) \text{ is } \hat{\gamma}_{MLE} = 1 - \Phi\left(\frac{1-\bar{x}}{S_n}\right) = \Phi\left(\frac{\bar{x}-1}{S_n}\right).$$

$$E_{\underline{\theta}}(x_1) = \mu, \quad E_{\underline{\theta}}(x_1^2) = \mu^2 + \sigma^2$$

$$\Rightarrow \hat{\mu}_{MLE} = \bar{x}, \quad \hat{\mu}_{MLE}^2 + \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2$$

$$\Rightarrow \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n x_i^2 - \bar{x}^2 = S_n^2$$

$$\Rightarrow \hat{\gamma}_{MLE} = \hat{\gamma}_{MLE} = \Phi\left(\frac{\bar{x}-1}{S_n}\right)$$

$$(iii) \ln L_2(\underline{\theta}) = -n \ln \Gamma(\alpha) - n\alpha \ln \sigma + (\alpha-1) \sum_{i=1}^n \ln x_i - \frac{1}{\sigma} \sum_{i=1}^n x_i, \quad -\infty < \mu < \infty, \quad \sigma > 0$$

$$\frac{\partial}{\partial \alpha} \ln L_2(\underline{\theta}) = -n \psi(\alpha) - n \ln \sigma + \sum_{i=1}^n \ln x_i$$

$$\frac{\partial}{\partial \sigma} \ln L_2(\underline{\theta}) = -\frac{n\alpha}{\sigma} + \frac{1}{\sigma^2} \sum_{i=1}^n x_i$$

$$\frac{\partial^2}{\partial \alpha^2} \ln L_2(\underline{\theta}) = -n \psi'(\alpha); \quad \frac{\partial^2}{\partial \sigma^2} \ln L_2(\underline{\theta}) = \frac{n\alpha}{\sigma^2} - \frac{2}{\sigma^3} \sum_{i=1}^n x_i$$

Critical points are solutions of the simultaneous equations

$$\psi(\hat{\alpha}_{MLE}) + \ln \hat{\sigma}_{MLE} = \frac{1}{n} \sum_{i=1}^n \ln x_i$$

$$\hat{\sigma}_{MLE} = \frac{1}{n \hat{\alpha}_{MLE}} \sum_{i=1}^n x_i$$

$$\Rightarrow \psi(\hat{\alpha}_{MLE}) - \ln \hat{\alpha}_{MLE} = \frac{1}{n} \sum_{i=1}^n \ln x_i - \bar{x} \quad \left(\text{note that } \frac{1}{n} \sum_{i=1}^n \ln x_i < \bar{x} \right)$$

$$\hat{\sigma}_{MLE} = \frac{\bar{x}}{n \hat{\alpha}_{MLE}}$$

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Let $R(x) = \psi(x) - \ln x$

$R'(x) = \psi'(x) - \frac{1}{x}$

According to theory of gamma functions $\psi'(x) > \frac{1}{x}, \forall x > 0$.

And $-\frac{1}{2} \leq \psi(x) - \ln x \leq -\frac{1}{2x}, \forall x > 0 \Rightarrow \psi(x) - \ln x$ takes all

Thus the Unique MLE exists

values between $(-\infty, 0)$

$E_0(X_1) = \alpha\sigma, \quad E_0(X_1^2) = \alpha(\alpha+1)\sigma^2$

$\Rightarrow \hat{\alpha}_{MLE} \hat{\sigma}_{MLE} = \bar{X}$ and $\hat{\alpha}_{MLE}(\hat{\alpha}_{MLE}+1) \hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$

$\Rightarrow (\hat{\alpha}_{MLE}, \hat{\sigma}_{MLE}) = \left(\frac{\bar{X}^2}{S_n^2}, \frac{S_n^2}{\bar{X}} \right)$

$\gamma(\alpha) = \text{Var}(X_1) = \alpha\sigma^2$

$\hat{\gamma}_{MLE} = \hat{\alpha}_{MLE} \hat{\sigma}_{MLE}^2$

$\hat{\gamma}_{MLE} = \hat{\alpha}_{MLE} \hat{\sigma}_{MLE}^2$

(iii) $\ln L_2(\theta) = -\ln 2 - n \ln \sigma - \frac{1}{\sigma} \sum_{i=1}^n |x_i - 1|$

$\rightarrow$ maximized at $\hat{\mu}_{MLE} = \text{median}\{x_1, \dots, x_n\}$ (may not be unique if n is even)
 $\hat{\sigma}_{MLE} = \frac{1}{n} \sum_{i=1}^n |x_i - 1|$

$\Rightarrow \hat{\mu}_{MLE} = \text{median}\{x_1, \dots, x_n\}$ (not unique if n is even)

$\hat{\sigma}_{MLE} = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{\mu}_{MLE}|$

$E_0(X_1) = \mu, \quad E_0\left(\left(X_1 - \frac{\mu}{\sigma}\right)^2\right) = E(Z^2) = 2 \quad \left(Z = \frac{X_1 - \mu}{\sigma} \sim \text{Exp}(0,1)\right)$

$\Rightarrow E_0(X_1^2) = \mu^2 + 2\sigma^2$

$\Rightarrow \hat{\mu}_{MLE} = \bar{X}, \quad \hat{\mu}_{MLE} + 2\hat{\sigma}_{MLE}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2$

$\Rightarrow (\hat{\mu}_{MLE}, \hat{\sigma}_{MLE}) = \left(\bar{X}, \frac{S_n}{\sqrt{2}} \right)$

Problem No. 5

For given data point x

$$\ln L_2(\theta) = \ln 4 - 4 \ln \theta + \ln(\theta - x_{(1)}) + \ln(\theta - x_{(2)}), \quad \theta > x_{(2)}$$

$$\begin{aligned} \frac{\partial}{\partial \theta} \ln L_2(\theta) &= -\frac{4}{\theta} + \frac{1}{\theta - x_{(1)}} + \frac{1}{\theta - x_{(2)}} \\ &= -\frac{2\theta^2 - 3\theta(x_{(1)} + x_{(2)}) + 4x_{(1)}x_{(2)}}{\theta(\theta - x_{(1)})(\theta - x_{(2)})}, \quad \theta > x_{(2)} \end{aligned}$$

Let $\hat{\theta}_1$ and $\hat{\theta}_2$ ($\hat{\theta}_1 < \hat{\theta}_2$) be roots of the equation, so that

$$\hat{\theta}_1 = \frac{3(x_{(1)} + x_{(2)}) - \sqrt{9x_{(1)}^2 + 9x_{(2)}^2 - 14x_{(1)}x_{(2)}}}{4}$$

$$\hat{\theta}_2 = \frac{3(x_{(1)} + x_{(2)}) + \sqrt{9x_{(1)}^2 + 9x_{(2)}^2 - 14x_{(1)}x_{(2)}}}{4}$$

and $\hat{\theta}_1 < x_{(2)} < \hat{\theta}_2$

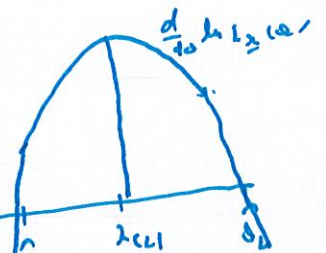
$\frac{\partial}{\partial \theta} \ln L_2(\theta) < (>) 0$ if $\theta > \hat{\theta}_2$ ($x_{(2)} < \theta < \hat{\theta}_2$)

$\Rightarrow L_2(\theta) \uparrow$ in $\theta \in (x_{(2)}, \hat{\theta}_2)$, and $\downarrow$ in $\theta \in (\hat{\theta}_2, \infty)$

$$\Rightarrow \hat{\theta}_{MLE} = \hat{\theta}_2 = \frac{3(x_{(1)} + x_{(2)}) + \sqrt{9(x_{(1)} - x_{(2)})^2 + 4x_{(1)}x_{(2)}}}{4}$$

$$E(x) = \frac{\theta}{3} \Rightarrow \hat{\theta}_{MLE} = 3\bar{x} = \frac{3(x_{(1)} + x_{(2)})}{2}$$

$\hat{\theta}_{MLE}$ and $\hat{\theta}_{MME}$ are functions of minimal sufficient statistic $\underline{T} = (x_{(1)}, x_{(2)})$.



Problem No. 6

For given data point x

$$(a) \ln L_2(\theta) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma_0^2) - \frac{1}{2\sigma_0^2} \sum_{i=1}^n (x_i - \theta)^2, \quad a < \theta < b$$

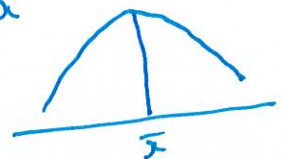
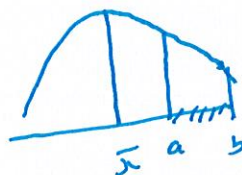
$$\frac{\partial}{\partial \theta} \ln L_2(\theta) = \frac{n}{\sigma_0^2} (\bar{x} - \theta) < (>) 0 \Leftrightarrow \theta > (<) \bar{x}$$

Cases

$\bar{x} < a$

$L_2(\theta) \downarrow$ on (a, b)

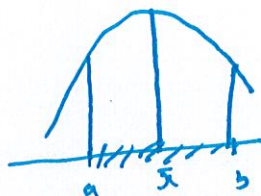
$$\Rightarrow \hat{\theta}_{MLE} = a$$



Case II $a \leq \bar{x} \leq b$

$L_2(\theta) \uparrow$ as $\theta \in (a, \bar{x})$, $\downarrow$ as $\theta \in (\bar{x}, b)$

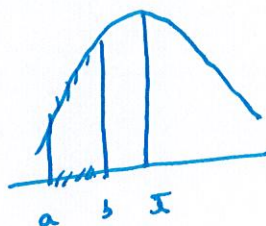
$$\Rightarrow \hat{\theta}_{MLE} = \bar{x}$$



Case III $\bar{x} > b$

$L_2(\theta) \uparrow$ as $\theta \in (a, b)$

$$\Rightarrow \hat{\theta}_{MLE} = b$$



Then,

$$\hat{\theta}_{MLE} = \begin{cases} a, & \bar{x} < a \\ \bar{x}, & a \leq \bar{x} \leq b \\ b, & \bar{x} > b \end{cases}$$

which is a function of minimal sufficient statistic $T = \bar{x}$; $\hat{\theta}_{MLE} = \hat{\theta}_{MNE} = \bar{x}$

(b) $L_2(\theta) = \frac{1}{\sigma_0^n} e^{-\frac{\sum x_i}{\sigma_0}} e^{\frac{n\theta}{\sigma_0}}$ $a < \theta < \min\{x_{(1)}, b\}$

$L_2(\theta)$ is $\uparrow$ as $\theta \in (a, \min\{x_{(1)}, b\})$
 $\Rightarrow \hat{\theta}_{MLE} = \min\{x_{(1)}, b\}$

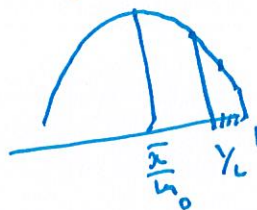
$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic $T = x_{(1)}$.

$$\hat{\theta}_{MNE} = \bar{x} - \sigma_0 \quad (E_0(x_i) = \theta + \sigma_0)$$

$\hat{\theta}_{MNE}$ is not a function of minimal sufficient statistic $T = x_{(1)}$.

(c) $\ln L_2(\theta) = \sum_{i=1}^n \ln\left(\frac{y_i}{x_i}\right) + n\bar{x} \ln \theta + n(\bar{x} - y_0)(1 - \theta)$ $\theta \in [\frac{1}{2}, 1]$

$$\frac{\partial}{\partial \theta} \ln L_2(\theta) = \frac{n\bar{x}}{\theta} + \frac{n(\bar{x} - y_0)}{1 - \theta} = n \frac{\bar{x} - y_0}{\theta(1 - \theta)} > (<) 0 \quad \text{if } \theta < (>) \frac{\bar{x}}{y_0}$$



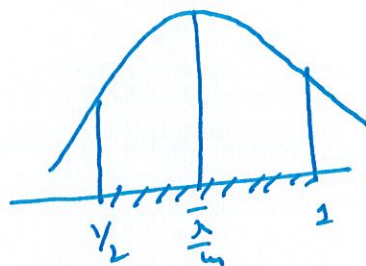
Case 5 $\frac{\bar{x}}{y_0} \leq \frac{1}{2}$

$L_2(\theta) \downarrow$ as $\theta \in [\frac{1}{2}, 1]$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{2}$$

Case II $\frac{\bar{x}}{m_0} > \frac{1}{2}$

$L_2(\theta) \uparrow$ as $\theta \in [\frac{1}{2}, \frac{\bar{x}}{m_0}]$ and
 $\downarrow$ as $\theta \in (\frac{\bar{x}}{m_0}, 1)$



$\Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{m_0}$

Thus $\hat{\theta}_{MLE} = \begin{cases} \frac{1}{2}, & \frac{\bar{x}}{m_0} < \frac{1}{2} \\ \frac{\bar{x}}{m_0}, & \frac{\bar{x}}{m_0} \geq \frac{1}{2} \end{cases} = \max \left\{ \frac{1}{2}, \frac{\bar{x}}{m_0} \right\}$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic $\bar{x}$.

$E_D(X_1) = m_0 \theta \Rightarrow \hat{\theta}_{MLE} = \frac{\bar{x}}{m_0}$, which is also a function of minimal sufficient statistic $\bar{x}$

(iv) $L_2(\underline{\theta}) = \frac{1}{(\theta_2 - \theta_1)^n}$, $\theta_1 \leq x_{(1)} \leq x_{(n)} \leq \theta_2$

is maximized at $(\hat{\theta}_1, \hat{\theta}_2) = (x_{(1)}, x_{(n)})$

$\Rightarrow \hat{\underline{\theta}}_{MLE} = (\hat{\theta}_{1,MLE}, \hat{\theta}_{2,MLE}) = (x_{(1)}, x_{(n)})$

$E_D(X_1) = \frac{\theta_1 + \theta_2}{2}$, $E_D(X_1^2) = \frac{\theta_1^2 + \theta_1 \theta_2 + \theta_2^2}{3}$

$\Rightarrow \frac{\hat{\theta}_{1,MLE} + \hat{\theta}_{2,MLE}}{2} = \bar{x}$, $\frac{\hat{\theta}_{1,MLE}^2 + \hat{\theta}_{1,MLE} \hat{\theta}_{2,MLE} + \hat{\theta}_{2,MLE}^2}{3} = \frac{1}{n} \sum_{i=1}^n X_i^2$

$\Rightarrow \hat{\theta}_{1,MLE} + \hat{\theta}_{2,MLE} = 2\bar{x} = \delta_1$ (say)

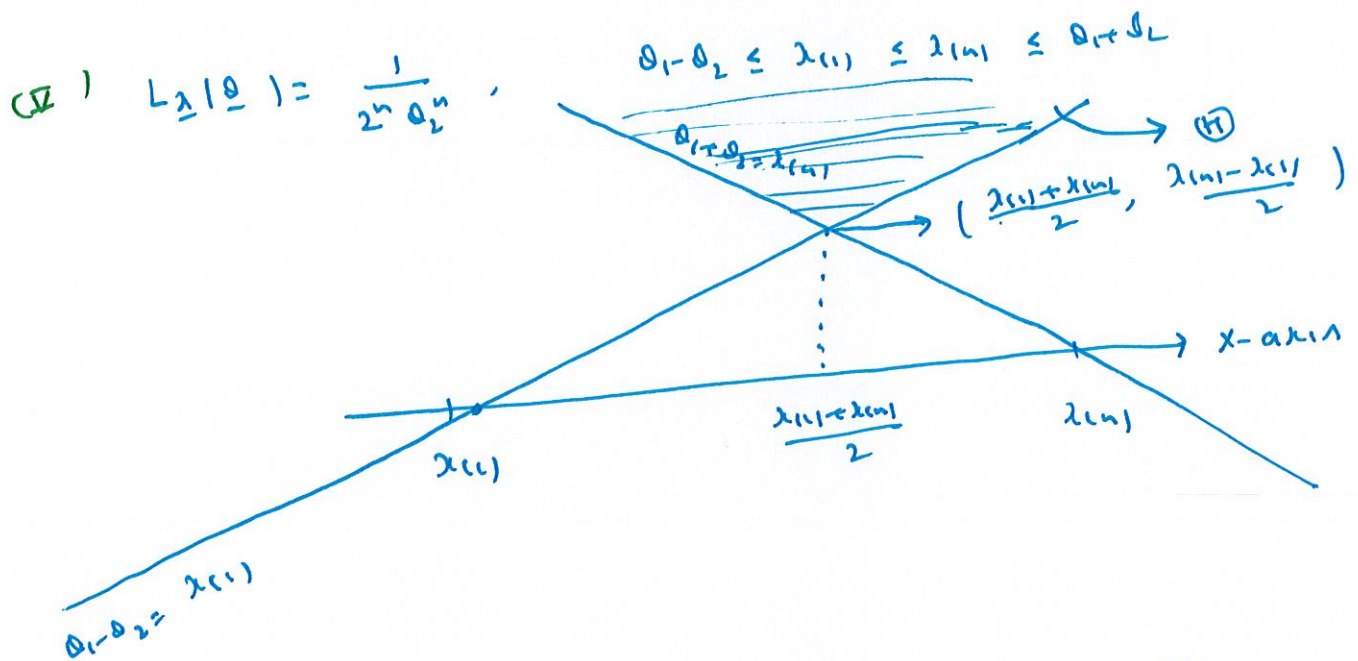
$\hat{\theta}_{1,MLE} \hat{\theta}_{2,MLE} = -\frac{3}{n} \sum_{i=1}^n X_i^2 + 4\bar{x}^2 = \delta_2$ (say)

$\Rightarrow \hat{\theta}_{1,MLE}^2 - \delta_1 \hat{\theta}_{1,MLE} + \delta_2 = 0$

Discriminant $\Delta = \delta_1^2 - 4\delta_2 = \frac{12}{n} \sum_{i=1}^n (X_i - \bar{x})^2 = 12S_n^2$

$\Rightarrow \hat{\theta}_{1,MLE} = \frac{\delta_1 - 2\sqrt{3}S_n}{2} = \bar{x} - \sqrt{3}S_n$, $\hat{\theta}_{2,MLE} = \bar{x} + \sqrt{3}S_n$
 $\Rightarrow \hat{\underline{\theta}}_{MLE} = (\hat{\theta}_{1,MLE}, \hat{\theta}_{2,MLE}) = (\bar{x} - \sqrt{3}S_n, \bar{x} + \sqrt{3}S_n)$

$\hat{\theta}_{MLE}$ is not a function of minimal sufficient statistic $I = (x_{(1)}, x_{(n)})$ unless $n=2$.



$L_2(\underline{\theta})$ is minimised where θ_2 is minimum. Thus

$$\hat{\underline{\theta}}_{MLE} = (\hat{\theta}_{1,MLE}, \hat{\theta}_{2,MLE}) = \left(\frac{x_{(1)} + x_{(n)}}{2}, \frac{x_{(n)} - x_{(1)}}{2} \right)$$

$$E_0(x_1) = \theta_1, \quad E_0(x_1^2) = \frac{(\theta_1 - \theta_2)^2 + (\theta_1 + \theta_2)(\theta_1 + \theta_2) + (\theta_1 + \theta_2)^2}{3} = \theta_1^2 + \frac{\theta_2^2}{3}$$

$$\Rightarrow \hat{\theta}_{1,MLE} = \bar{X}, \quad \hat{\theta}_{1,MLE}^2 + \frac{\hat{\theta}_{2,MLE}^2}{3} = \frac{1}{n} \sum_{i=1}^n x_i^2$$

$$\Rightarrow \hat{\theta}_{2,MLE}^2 = 3 S_n^2 \quad (S_n^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{X})^2)$$

$$\Rightarrow \hat{\underline{\theta}}_{MLE} = (\hat{\theta}_{1,MLE}, \hat{\theta}_{2,MLE}) = (\bar{X}, \sqrt{3} S_n)$$

$\hat{\theta}_{MLE}$ is a function of minimal sufficient statistic $I = (x_{(1)}, x_{(n)})$
 $\hat{\theta}_{MLE}$ is not a function of minimal sufficient statistic $I = (x_{(1)}, x_{(n)})$ unless $n=2$.

Problem No. 7

$$P_0(x_1 > c) = e^{-\frac{c}{\theta}}$$

For given data point $y = (y_1, \dots, y_n)$, the likelihood function is

$$L_2(\theta) = \left(\prod_{j=1}^n \frac{1}{\theta} e^{-\frac{y_j}{\theta}} \right) \left(e^{-\frac{c}{\theta}} \right)^{n-m}, \quad \theta > 0$$

$$\ln L_2(\theta) = -n \ln \theta - \frac{m\bar{y}}{\theta} - (n-m)\frac{c}{\theta}, \quad \theta > 0$$

$$\frac{\partial}{\partial \theta} \ln L_2(\theta) = -\frac{n}{\theta} + \frac{m\bar{y}}{\theta^2} + \frac{(n-m)c}{\theta^2} > (<) 0$$

$$\Leftrightarrow \theta < (>) \frac{m\bar{y} + (n-m)c}{n}$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{m\bar{y} + (n-m)c}{n} = \bar{y} + \left(\frac{n}{m} - 1\right)c.$$

Problem No. 8

We will use the Jensen inequality which states that for any non-degenerate r.v. x and any convex function $h(\cdot)$

$$E(h(x)) > h(E(x))$$

$$h_1(x) = e^x \text{ is strictly convex on } \mathbb{R} \Rightarrow E(h_1(x)) > h_1(E(x))$$

$$\Rightarrow E(e^x) > e^{E(x)}$$

$$h_2(x) = \frac{1}{x} \text{ is strictly convex on } (0, \infty) \Rightarrow E(h_2(x)) > h_2(E(x))$$

$$\Rightarrow E\left(\frac{1}{x}\right) > \frac{1}{E(x)}$$

$$\Rightarrow E\left(x \cdot E\left(\frac{1}{x}\right)\right) > 1$$

$$h_3(x) = -\ln x \text{ is strictly convex on } (0, \infty) \Rightarrow E(h_3(x)) > h_3(E(x))$$

$$\Rightarrow E(-\ln x) > -\ln E(x)$$

$$\Rightarrow E(\ln x) < \ln E(x)$$

Consider a r.v. x having p.m.b.

$$p_x(x) = \begin{cases} w_i, & \text{if } x = a_i, i=1, \dots, n \\ 0, & \text{otherwise} \end{cases}$$

Since $w_i > 0, i=1, \dots, n$, and $\sum_{i=1}^n w_i = 1$, $p_x(\cdot)$ is a proper p.m.b.

$$E(\ln x) < \ln E(x) \Rightarrow \sum_{i=1}^n w_i \ln a_i < \ln \left(\sum_{i=1}^n a_i w_i \right)$$

$$\Rightarrow \ln \left(\prod_{i=1}^n a_i^{w_i} \right) < \ln \left(\sum_{i=1}^n a_i w_i \right) \Rightarrow \prod_{i=1}^n a_i^{w_i} < \sum_{i=1}^n a_i w_i$$

... (*)

Also

$$E(\ln \frac{1}{x}) < \ln E(\frac{1}{x}) \quad (\text{using } \frac{1}{x} \text{ in place of } x \text{ in (iii)})$$

$$\Rightarrow -E(\ln x) < \ln E(\frac{1}{x})$$

$$\Rightarrow -\sum_{i=1}^n (\ln a_i) \omega_i < \ln \left(\sum_{i=1}^n \frac{\omega_i}{a_i} \right)$$

$$\Rightarrow -\ln \left(\prod_{i=1}^n a_i^{\omega_i} \right) < \ln \left(\sum_{i=1}^n \frac{\omega_i}{a_i} \right)$$

$$\Rightarrow \ln \left(\frac{1}{\prod_{i=1}^n a_i^{\omega_i}} \right) < \ln \left(\sum_{i=1}^n \frac{\omega_i}{a_i} \right)$$

$$\Rightarrow \frac{1}{\prod_{i=1}^n a_i^{\omega_i}} < \sum_{i=1}^n \frac{\omega_i}{a_i} \quad \dots \quad (\text{II})$$

Combining (I) and (II), we get (Arithmetic inequality)

$$\sum_{i=1}^n a_i \omega_i > \prod_{i=1}^n a_i^{\omega_i} > \frac{1}{\sum_{i=1}^n \frac{\omega_i}{a_i}}$$

Problem No. 9

$\delta(x)$ has p.d.b. $p_x(a) = \phi(a - \frac{x_1+x_2}{2})$
 (i) $\bar{x}$ is ^{complete-sufficient and hence} minimal sufficient. The derived ^{randomized} estimator $\delta^*(x)$ has
 the p.d.b. $p_t^*(a)$, where

$$p_t^*(a) = E \left[p_x(a) \mid \bar{x} = t \right]$$

$$= E \left[\phi \left(a - \frac{x_1+x_2}{2} \right) \mid \bar{x} = t \right]$$

$$= E \left[\phi \left(a - \left(\frac{x_1+x_2}{2} - \bar{x} \right) - t \right) \mid \bar{x} = t \right]$$

$$= E \left[\phi \left(a - \left(\frac{x_1+x_2}{2} - \bar{x} \right) - t \right) \right] \quad (\text{Baru's Theorem})$$

$$= E \left[\phi \left(\frac{x_1+x_2}{2} - \bar{x} + t - a \right) \right]$$

$$= \int_{-\infty}^{\infty} \phi(b+t-a) \sqrt{\frac{2n}{n-2}} \phi \left(\sqrt{\frac{2n}{n-2}} z \right) dz \quad \left(\frac{x_1+x_2}{2} - \bar{x} \sim N(0, \frac{n-2}{2n}) \right)$$

$$= \frac{\sqrt{\frac{2n}{n-2}}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}(b+t-a)^2} e^{-\frac{n}{n-2} z^2} dz$$

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$$\frac{1}{2} (z + t - a)^2 + \frac{n}{n-2} z^2 = \frac{1}{2} \left[\frac{3n-2}{n-2} z^2 + 2(t-a)z + (t-a)^2 \right]$$

$$= \frac{3n-2}{2(n-2)} \left[z^2 + \frac{2(n-2)}{3n-2} (t-a)z + \frac{n-2}{3n-2} (t-a)^2 \right]$$

$$= \frac{3n-2}{2(n-2)} \left[\left\{ z + \frac{n-2}{3n-2} (t-a) \right\}^2 - \frac{(n-2)^2}{(3n-2)^2} (t-a)^2 + \frac{n-2}{3n-2} (t-a)^2 \right]$$

$$= \frac{3n-2}{2(n-2)} \left(z + \frac{n-2}{3n-2} (t-a) \right)^2 + \frac{n}{3n-2} (t-a)^2$$

$$p_z^*(a) = \frac{\sqrt{\frac{2n}{n-2}}}{2\pi} e^{-\frac{n}{3n-2} (t-a)^2} \int_{-\infty}^{\infty} e^{-\frac{3n-2}{2(n-2)} \left(z + \frac{n-2}{3n-2} (t-a) \right)^2} dz$$

$$= \frac{\sqrt{\frac{2n}{n-2}}}{2\pi} \times e^{-\frac{n}{3n-2} (t-a)^2} \times \sqrt{2\pi} \times \sqrt{\frac{n-2}{3n-2}}$$

$$= \sqrt{\frac{2n}{3n-2}} \phi \left(\sqrt{\frac{2n}{3n-2}} (a-t) \right)$$

$$\Rightarrow \delta^* \sim N \left(\bar{x}, \frac{3n-2}{2n} \right)$$

(ii) A non-randomized estimator better than δ^* and based on minimal sufficient statistic $\bar{x}$ is

$$\delta_0(\underline{x}) = E^a[\delta^*(\underline{x})] = \bar{x}.$$

Problem No. 10

$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ is complete-sufficient (and hence minimal sufficient).

(i) $\delta(\underline{x}) \sim U\left(\frac{n-1}{n} \bar{x}_{n-1}, \bar{x}_{n-1}\right)$ has p.d.f.

$$p_{\underline{x}}(a) = \frac{n}{\bar{x}_{n-1}} \mathbb{I} \left(\frac{n-1}{n} \bar{x}_{n-1} < a < \bar{x}_{n-1} \right)$$

The derived ^{randomized} estimator $\delta^*(\underline{x})$ has p.d.f. (for $\bar{x} = t$)

$$p_t^*(a) = E \left[\frac{n}{\bar{x}_{n-1}} I \left(\frac{n-1}{n} \bar{x}_{n-1} \leq a \leq \bar{x}_{n-1} \right) \mid \bar{x} = t \right]$$

$$= \frac{n}{t} E \left[\frac{\bar{x}}{\bar{x}_{n-1}} I \left(\frac{a}{t} \leq \frac{\bar{x}_{n-1}}{\bar{x}} \leq \frac{n}{n-1} \frac{a}{t} \right) \mid \bar{x} = t \right]$$

$$= \frac{n}{t} E \left[\frac{n-1}{n \bar{z}} I \left(\frac{(n-1)a}{n t} < \bar{z} < \frac{a}{t} \right) \right] \quad (\text{Darwin's Thm}),$$

Where $\bar{z} = \frac{\sum_{i=1}^{n-1} x_i}{\sum_{i=1}^n x_i} = \frac{\sum_{i=1}^{n-1} x_i}{\sum_{i=1}^{n-1} x_i + x_n} \sim \text{Beta}(n-1, 1)$ has pdf $b_z(z) = (n-1) z^{n-2}$, $0 < z < 1$

$$p_t^*(a) = \frac{n-1}{t} E \left[\frac{1}{\bar{z}} I \left(\frac{(n-1)a}{n t} < \bar{z} < \frac{a}{t} \right) \right]$$

$$a < 0, \quad \text{or} \quad a \geq \frac{n t}{n-1}$$

$$= \begin{cases} 0, & a/t \\ \frac{n-1}{t} \int_{\frac{(n-1)a}{n t}}^{\frac{a}{t}} \frac{1}{z} \times (n-1) z^{n-2} dz & 0 \leq a < t \\ \frac{n-1}{t} \int_{\frac{(n-1)a}{n t}}^1 \frac{1}{z} \times (n-1) z^{n-2} dz & t \leq a < \frac{n}{n-1} t \end{cases}$$

$$a < 0 \quad \text{or} \quad a \geq \frac{n t}{n-1}$$

$$= \begin{cases} 0, & 0 \leq a < t \\ \frac{(n-1)^2}{(n-2) t^{n-1}} \left[1 - \left(\frac{n-1}{n} \right)^{n-1} \right] a^{n-2}, & 0 \leq a < t \\ \frac{(n-1)^2}{(n-2) t} \left[1 - \left(\frac{n-1}{n} \right)^{n-1} \frac{a^{n-2}}{t^{n-2}} \right], & t \leq a \leq \frac{n}{n-1} t \end{cases}$$

(iii) The derived non-randomized estimator $\hat{S}^*(x) = E(S^*(x))$

$$\hat{S}^*(x) = \frac{(n-1)^2}{(n-2) \bar{x}^{n-1}} \left[1 - \left(\frac{n-1}{n} \right)^{n-1} \right] \int_0^{\bar{x}} a^{n-1} da$$

$$+ \frac{(n-1)^2}{(n-2) \bar{x}} \int_{\frac{n-1}{n} \bar{x}}^{\frac{n}{n-1} \bar{x}} a \left[1 - \left(\frac{n-1}{n} \right)^{n-1} \frac{a^{n-2}}{\bar{x}^{n-2}} \right] da$$

$$= \frac{2n-1}{2n} \bar{x}$$

Problem No. 11

(i) $T = \bar{X}$ is Complete-Sufficient and hence minimal Sufficient. For any convex loss function $L(\theta, d)$ (in particular absolute error and SEL) convex w.r.t. θ , the derived estimator is

$$S_1^*(x) = E(S_1(x) | \bar{x})$$

$$= \sum_{i=1}^n \alpha_i E(x_i | \bar{x})$$

Since $(x_1, \dots, x_n) \stackrel{d}{=} (x_{k_1}, \dots, x_{k_n})$ for every permutation $(k_1, \dots, k_n)$ of $(1, \dots, n)$

$$E(x_i | \bar{x}) = E(x_1 | \bar{x}) = c, \text{ say } (c \text{ does not depend on } \theta \text{ as } \bar{x} \text{ is Sufficient})$$

$$\Rightarrow \sum_{i=1}^n E(x_i | \bar{x}) = nc$$

$$\Rightarrow E\left(\sum_{i=1}^n x_i | \bar{x}\right) = nc \Rightarrow E(n\bar{x} | \bar{x}) = nc \Rightarrow n\bar{x} = nc$$

$$\Rightarrow E(x_i | \bar{x}) = c, \bar{x}, i=1, \dots, n,$$

$$\Rightarrow S_1^*(x) = \bar{x} \left(\sum_{i=1}^n \alpha_i \right) = \bar{x}.$$

(ii) $S_2(x) = x_{(n+1)}$ is degenerate randomized estimator, i.e. $S_2(x)$ has p.m.b. $p_2(a) = I(a = x_{(n+1)})$. The derived randomized estimator S_2^* has p.m.b.

$$p_k^*(a) = E(I(a = x_{(n+1)}) | \bar{x} = t)$$

$$= P(a = x_{(n+1)} | \bar{x} = t)$$

$$= P(a = x_{(n+1)} - \bar{x} + t | \bar{x} = t)$$

$$= P(a = x_{(n+1)} - \bar{x} + t) \quad (\text{Bayes's Theorem})$$

Let $Z_t \stackrel{d}{=} x_{(n+1)} - \bar{x} + t$. Clearly the dist. of Z_t does not depend on θ . Th $Z_t \sim p_k^*(a)$

(iii) The derived non-randomized estimator is

$$\hat{S}_2(X) = E(Z_{\bar{X}})$$

$$E(Z_t) = E(X_{(m+1)} - \bar{X} + t) \\ = E(X_{(m+1)} - \bar{X}) + t$$

$$(X_1 - \mu, \dots, X_n - \mu) \stackrel{d}{=} (\mu - X_1, \dots, \mu - X_n)$$

$$\Rightarrow (X_{(m+1)} - \mu) - (\bar{X} - \mu) \stackrel{d}{=} (\mu - X_{(m+1)}) - (\mu - \bar{X})$$

$$\Rightarrow (X_{(m+1)} - \bar{X}) \stackrel{d}{=} (\bar{X} - X_{(m+1)})$$

$$\Rightarrow E(X_{(m+1)} - \bar{X}) = E(\bar{X} - X_{(m+1)})$$

$$\Rightarrow E(Z_t) = t$$

Then the derived estimator is $\hat{S}_2(X) = \bar{X}$.

Problem 11.12

The derived non-randomized estimator is $S^*(X)$

$$= E(S(X))$$

$$= \frac{X}{n} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2}$$

$$= \frac{X}{2n} + \frac{1}{4}$$

$$L^*(\theta, S(X)) = \frac{1}{2} \left(\frac{X}{n} - \theta \right)^2 + \frac{1}{2} \left(\frac{1}{2} - \theta \right)^2$$

$$R(\theta, S) = E_{\theta} [L^*(\theta, S(X))]$$

$$= \frac{1}{2} E_{\theta} \left[\left(\frac{X}{n} - \theta \right)^2 \right] + \frac{1}{2} \left(\frac{1}{2} - \theta \right)^2$$

$$= \frac{\theta(1-\theta)}{2n} + \frac{(1-2\theta)^2}{8}, \quad 0 < \theta < 1$$

$$R(\theta, S^*) = E_{\theta} \left[\left(\frac{X}{2n} + \frac{1}{4} - \theta \right)^2 \right] = V \left(\frac{X}{2n} + \frac{1}{4} - \theta \right) + \left(E \left[\left(\frac{X}{2n} + \frac{1}{4} - \theta \right) \right] \right)^2 \\ = \frac{\theta(1-\theta)}{4n} + \frac{(1-2\theta)^2}{16}, \quad 0 < \theta < 1$$

Clearly,

$$R(\theta, \theta') < R(\theta, \theta), \quad \forall \theta \in (0, 1)$$

Problem No. 13

$$L(\theta, a) = |a - \theta|$$

$T(X) = X_{(n)}$ is complete-sufficient, and hence minimal sufficient.

(i) The derived estimator is $\eta_0(T)$, where

$$\eta_0(t) = E(\delta_0(X) | X_{(n)} = t)$$

$$= E(\bar{X} | X_{(n)} = t)$$

$$= t E\left(\frac{\bar{X}}{X_{(n)}} | X_{(n)} = t\right)$$

$$= t E\left(\frac{\bar{X}}{X_{(n)}}\right) \quad (\text{By Basu's Theorem as } \frac{\bar{X}}{X_{(n)}} \text{ is ancillary and } X_{(n)} \text{ is c.s.})$$

Again, by Basu's Theorem,

$$E(\bar{X}) = E(X_{(n)} \frac{\bar{X}}{X_{(n)}}) = E(X_{(n)}) E\left(\frac{\bar{X}}{X_{(n)}}\right)$$

$$\Rightarrow E\left(\frac{\bar{X}}{X_{(n)}}\right) = \frac{E(\bar{X})}{E(X_{(n)})} = \frac{\theta/2}{\int_0^\theta t \times \frac{n+1}{\theta^n} dt} = \frac{n+1}{2n}$$

$$\Rightarrow \eta_0(T) = \frac{n+1}{2n} X_{(n)}$$

dominates the estimator $\delta_0(X)$ under any convex loss function $L(\theta, a)$ (convex in a) and hence, in particular, under absolute error.

(ii) The derived estimator is $\eta_1(T) = E(\delta_1(X) | T)$. We have

$$\eta_1(t) = E_0(\delta_1(X) | T=t) = E_0(X_{(n-1)} | X_{(n)} = t) = t E\left(\frac{X_{(n-1)}}{X_{(n)}} | X_{(n)} = t\right)$$

$$= t E\left(\frac{X_{(n-1)}}{X_{(n)}}\right) \quad (\text{By Basu's Thm})$$

$$\text{Also, by Basu's Thm, } E(X_{(n-1)}) = E(X_{(n)} \times \frac{X_{(n-1)}}{X_{(n)}}) = E(X_{(n)}) E\left(\frac{X_{(n-1)}}{X_{(n)}}\right)$$

$$\Rightarrow E\left(\frac{X_{(n-1)}}{X_{(n)}}\right) = \frac{E(X_{(n-1)})}{E(X_{(n)})} = \frac{\int_0^1 n(n-1)x^{n-2} \times t^{n-1} dt}{\int_0^1 t \times n t^{n-1} dt} = \frac{n-1}{n}$$

$$\Rightarrow \eta_1(T) = \frac{n-1}{n} X_{(n)} \text{ dominates } \delta_1(X) = X_{(n-1)}.$$

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